Approximate Functions

VLC = Vision Learning Control

Going Deep

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Overview

- No free lunch and universal approximation
- Why go deep?
- Problems of going deep
- Some fixes:
  - Improving gradient flow with skip connections
  - Regularising with Dropout

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  - **No machine learning algorithm is universally better than any other!**
  - Fortunately, in the real world, data is generated by a small subset of generating distributions...

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$\implies$ simple neural networks can represent a wide variety of interesting functions when given appropriate parameters.

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    - The optimiser might not find a good solution[2].
    - The training algorithm might just choose the wrong solution as a result of overfitting.
    - *There is no known universal procedure for examining a set of examples and choosing a function that will generalise to points out of the training set.*

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Then Why Go Deep?

- There are functions you can compute with a deep neural network that shallow networks require exponentially more hidden units to compute.
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- Alternatively, one could just consider that a deep architecture just expresses that the function we wish to learn is a program made of multiple steps where each step makes use of the previous steps outputs.
- **Empirically, deeper networks just seem to generalise better!**

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  - Problems of gradient flow
  - Horrible symmetries in the loss landscape
  - Overfitting

Vanishing and Exploding Gradients

- The vanishing and exploding gradient problem is a difficulty found in training NN with gradient-based learning methods and backpropagation.
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- Adaptive optimisers can rescale small parameter gradients, but they cannot recover information lost through repeated poorly conditioned Jacobians.

Residual Connections

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- This is not the full story: skip connections also change the optimisation geometry and symmetries.

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Residual Connections

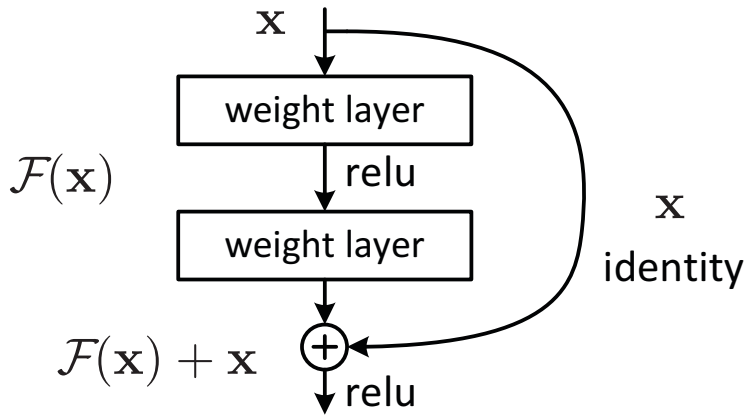


Figure 2. Residual learning: a building block.

K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," CVPR, Las Vegas, NV, 2016, pp. 770-778.

Residual Gradient Flow

A residual block computes

$$\boldsymbol{y} = \boldsymbol{x} + F(\boldsymbol{x}; \boldsymbol{\theta}) \, .$$

If $J_F(\boldsymbol{x})$ denotes the Jacobian of F evaluated at $\boldsymbol{x}$, then the Jacobian of the block is

$$\frac{\partial \boldsymbol{y}}{\partial \boldsymbol{x}} = I + J_F(\boldsymbol{x}).$$

For a scalar loss L and column-vector gradients, backprop computes $\nabla_{\boldsymbol{\theta}} L$ to update the parameters and $\nabla_{\boldsymbol{x}} L$ to propagate the gradient to earlier layers:

$$\nabla_{\boldsymbol{x}} L = \nabla_{\boldsymbol{y}} L + \underbrace{J_F(\boldsymbol{x})^T \nabla_{\boldsymbol{y}} L}_{\text{VJP through } F} .$$

- The gradient has a direct identity path as well as the path through F.
- This improves gradient flow, although it does not guarantee that gradients remain well scaled.

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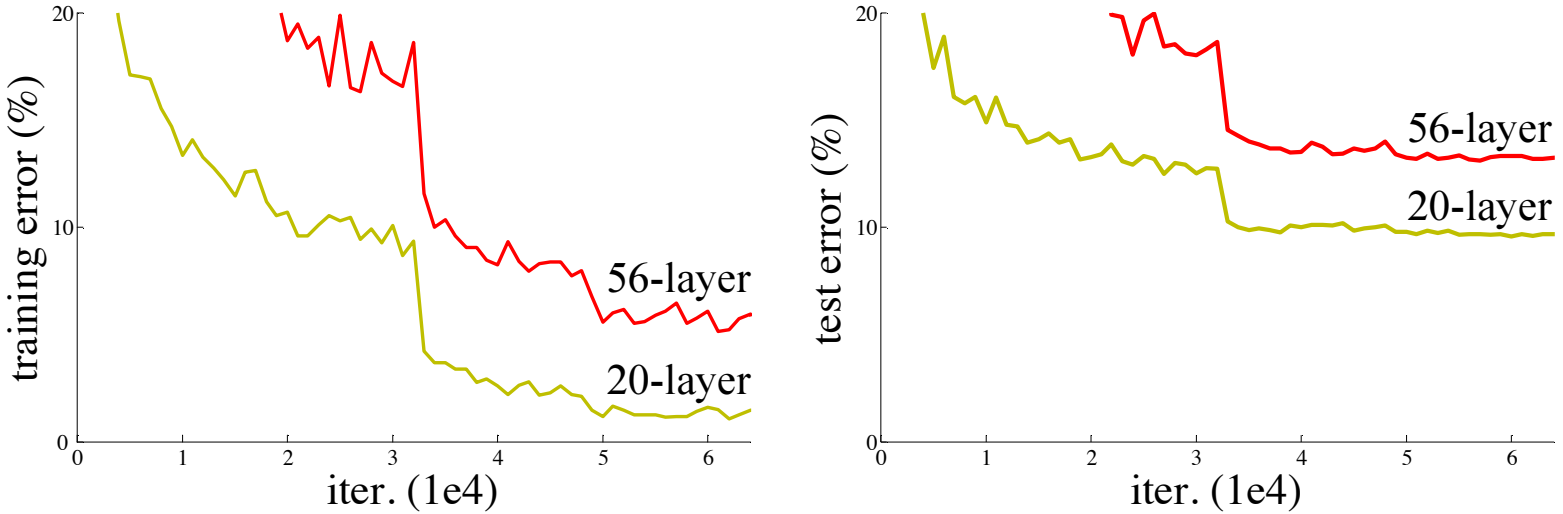


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer "plain" networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

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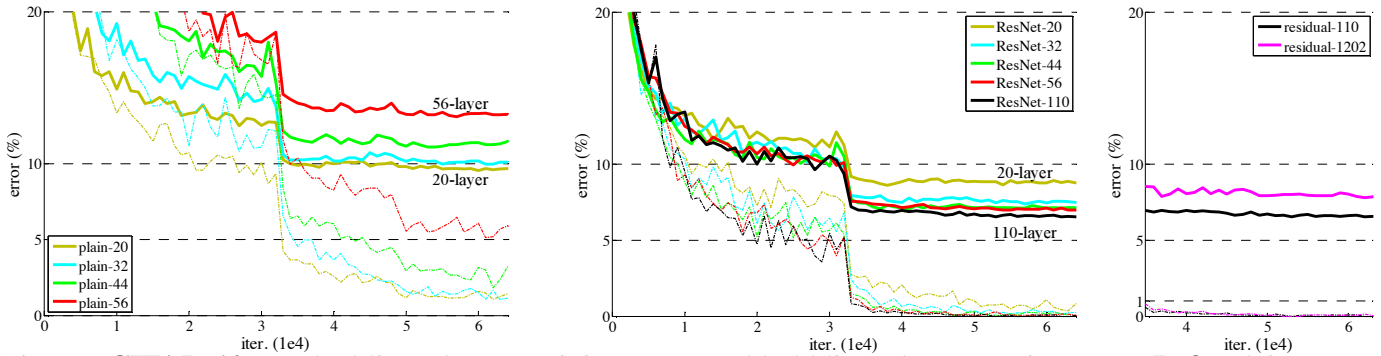


Figure 6. Training on **CIFAR-10**. Dashed lines denote training error, and bold lines denote testing error. **Left**: plain networks. The error of plain-110 is higher than 60% and not displayed. **Middle**: ResNets. **Right**: ResNets with 110 and 1202 layers.

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  - Use a stochastic weight regulariser (like dropout)

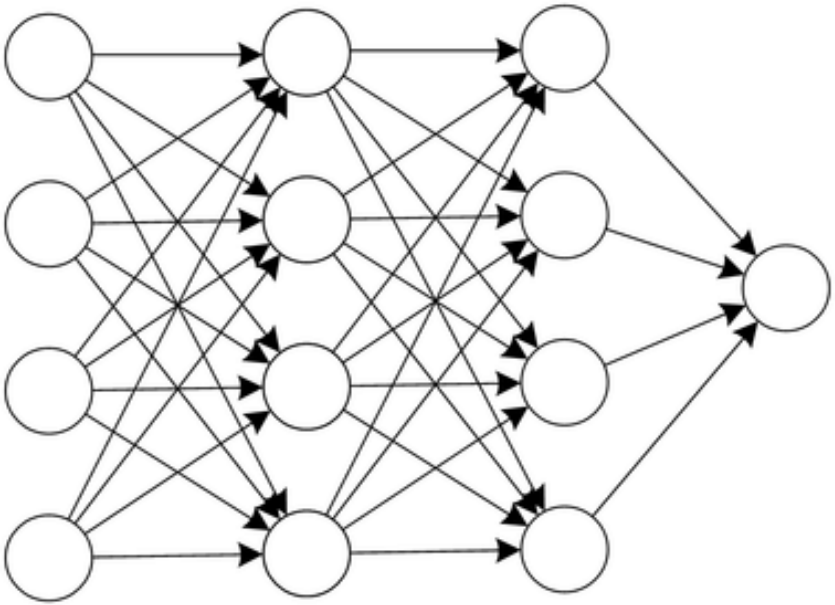
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- The key idea in dropout is to randomly drop neurons, including all of the connections, from the neural network during training.

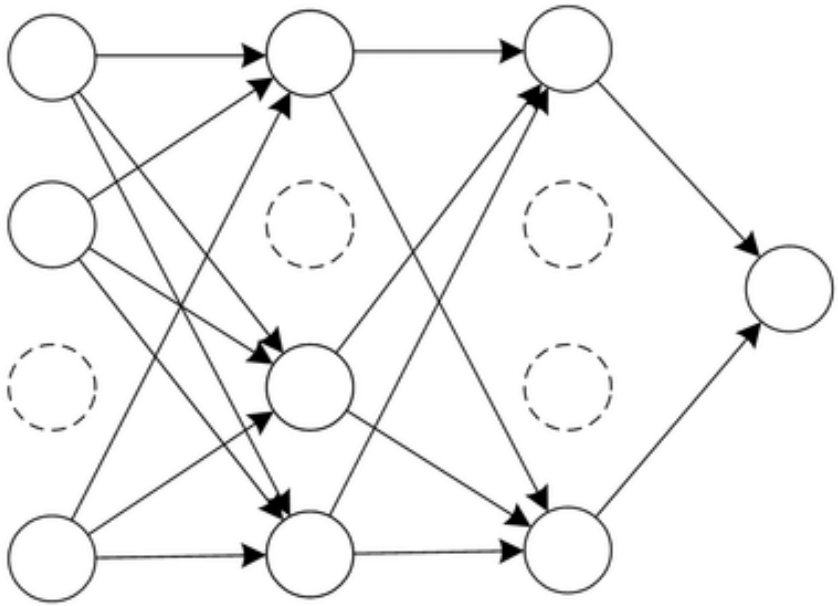
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- Motivation: averaging predictions from several models often improves generalisation.
  - Training many networks independently is expensive.
  - Dropout samples many overlapping subnetworks that share parameters.

Dropout



(a) Standard Neural Network



(b) Network after Dropout

Image from: `https://www.researchgate.net/figure/Dropout-neural-network-model-a-is-a-standard-neural-network-b-is-the-same-network_fig3_309206911`

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- On each training forward pass, we sample a binary mask and apply it to the activations.
- **Inverted Dropout** rescales the retained activations so that their expected magnitude is unchanged when dropout is disabled at evaluation/test time.

How is Inverted Dropout implemented?

- We define a random binary mask $\boldsymbol{m}^{(l)}$ for the activations in layer l. Each element is sampled from a Bernoulli distribution with $P(m_i^{(l)} = 1) = p$, where p is the *keep probability*. A new mask is sampled on each training forward pass.

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- The forward pass of a Dropout layer (function) during **training** is given by $f(\boldsymbol{x}) = \boldsymbol{x} \odot \boldsymbol{m}/p$.
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- The gradient (during training) is simply the hadamard product of the incoming gradient with $\boldsymbol{m}/p$.

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